

Teacher Answer Key:

Mechanical Oscillations – Ex 5: Damped vs Undamped

1. Initial Conditions ($t = 0$)

Figure (a) & (b): Both graphs start from the maximum displacement ($x = 5$ cm) with a horizontal tangent (slope = 0).

Conditions for both situations: $x_0 = 5$ cm, $v_0 = 0$

2. Study of Figure (a) (Undamped)

a) Justification:

The amplitude of oscillations (X_m) remains constant at 5 cm over time. This indicates that the Mechanical Energy (ME) is conserved and friction is negligible. Thus, the oscillations are free and **un-damped**.

b) Differential Equation:

Conservation of Mechanical Energy ($ME = KE + EPE$):

$$\frac{dME}{dt} = 0 \implies mx'' + Kx = 0$$

Result: $x'' + \frac{K}{m}x = 0$

This is a second-order linear differential equation with constant coefficients, characteristic of Simple Harmonic Motion (SHM) of the form $x'' + \omega_0^2 x = 0$.

c) Expression of Period T_0 :

From the differential equation form $x'' + \omega_0^2 x = 0$, we identify $\omega_0^2 = \frac{K}{m}$.

Since $T_0 = \frac{2\pi}{\omega_0}$, the proper period is given by:

Expression:

$$T_0 = 2\pi\sqrt{\frac{m}{K}}$$

This period is called the **Proper (or Natural) Period**.

d) Mass Calculation:

Reading from the graph, the proper period is: $T_0 \approx 3.4$ s

Using the period formula: $T_0^2 = 4\pi^2 \frac{m}{K} \implies m = \frac{KT_0^2}{4\pi^2}$

Given $K = 85$ N/m and taking $\pi^2 \approx 10$:

$$m = \frac{85 \times (3.4)^2}{40} = \frac{85 \times 11.56}{40}$$

Calculation: $85 \times 11.56 = 982.6$. Then $982.6/40 = 24.565$.

Result: $m \approx 24.6$ kg

3. Study of Figure (b) (Damped)

a) Justification:

The amplitude of the oscillations decreases gradually with time. This indicates the presence of friction (non-conservative forces) dissipating the mechanical energy.

b) Duration T' :

The duration separating two successive maxima is called the **Pseudo-Period**.

Reading from graph: $T' \approx 3.6 \text{ s}$

Note: The pseudo-period (3.6 s) is slightly longer than the proper period (3.4 s) due to damping.

c) Energy Lost in $[0; T']$:

We calculate the Mechanical Energy variation $\Delta ME = ME_f - ME_i$.

At $t = 0$: $x_0 = 5 \text{ cm} = 0.05 \text{ m}$, $v_0 = 0$.

$$ME_0 = EPE_{max} = \frac{1}{2}Kx_0^2 = \frac{1}{2}(85)(0.05)^2$$

$$ME_0 = 42.5 \times 0.0025 = 0.106 \text{ J}$$

At $t = T' = 3.6 \text{ s}$: Assuming the amplitude decays to approximately $x_1 \approx 3.5 \text{ cm} = 0.035 \text{ m}$ (estimate from typical decay).

$$ME_1 = \frac{1}{2}Kx_1^2 = \frac{1}{2}(85)(0.035)^2$$

$$ME_1 = 42.5 \times 0.001225 \approx 0.052 \text{ J}$$

Energy Lost:

$$|\Delta ME| = 0.106 - 0.052 = 0.054 \text{ J}$$

Result: $E_{lost} \approx 5.4 \times 10^{-2} \text{ J}$

d) Average Power:

$$P_{avg} = \frac{|\Delta ME|}{T'} = \frac{0.054}{3.6}$$

Result: $P_{avg} \approx 0.015 \text{ W}$ (or 15 mW)

e) Period of Driven Oscillations:

To maintain the oscillations (compensate for energy loss) at resonance, the period of the external exciter (T_{driven}) must be equal to the natural period of the oscillator.

Result: $T_{driven} = T_0 = 3.4 \text{ s}$